

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. CHB-603: Organic Chemistry-IV

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 which is compulsory.

Question 1. Answer all parts of the following: [5 × 2 = 10]

- (a) Discuss the synthesis and uses of sulphanilamide.
- (b) Discuss the synthesis of chloroquine.
- (c) Write down the monomers of Nylon-6,6 and Bakelite.
- (d) Differentiate between fibrous and globular proteins.
- (e) What is the zwitterionic nature of amino acids?

Question 2.

- (a) Describe Darapsky and Strecker synthesis of amino acids. [7]
- (b) Write one method for the synthesis of each of the following amino acids: (i) Proline, (ii) Cysteine, and (iii) Aspartic acid. [7]

Question 3. Write short notes on the following pharmaceutical compounds: [14]

- (a) Chloramphenicol (structure, synthesis, and uses)
- (b) Aspirin (synthesis, mechanism of action, and uses)

Question 4.

- (a) Discuss the synthesis of Citral and outline its structure elucidation. [8]
- (b) What are the differences between DNA and RNA? Discuss their secondary structures. [6]

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